Agile Software Development and DevOps Lab

Experiment 4

Aim: To understand Continuous Integration, install Jenkins, and build CI/CD pipelines using Declarative and Scripted syntax.

LO mapped: LO

LO:

Theory:

Continuous Integration

Continuous Integration is a software development practice that promotes collaboration, early issue detection, and automated testing to enhance the quality and efficiency of software development.

What is Jenkins?

Jenkins is an industry-standard, open-source automation server designed to facilitate Continuous Integration (CI) and Continuous Delivery/Deployment (CD) in modern software engineering. It serves as the central orchestration engine for modern DevOps practice, replacing manual build and deployment steps with automated, repeatable workflows.

Key Functional Areas:

- Software Lifecycle Automation: Jenkins automates critical development stages—including code compilation, static analysis, unit testing, artifact generation, environment provisioning, and production deployment.
- Early Issue Detection (CI): Every time developers commit code changes to a shared repository (e.g., GitHub or GitLab), Jenkins can automatically trigger a build and execute test suites. This provides immediate feedback on integration breaks or bugs early in the cycle.
- Extensible Architecture & Plugins: With an ecosystem of over 1,800 community-contributed plugins, Jenkins integrates seamlessly with nearly every tool in the software stack—including version control systems (Git, SVN), build tools (Maven, Gradle), container platforms (Docker, Kubernetes), and cloud infrastructure (AWS, Azure, GCP).
- Pipeline as Code: Jenkins allows teams to define complex, multi-stage workflows as code using Jenkinsfiles (written in Groovy). This ensures build logic is version-controlled, easily audit-able, and consistent across development, staging, and production environments.

What is a Jenkins Pipeline?

A Jenkins Pipeline is a set of automated processes that help developers and DevOps teams build, test, and deploy code in a CI/CD environment. In simple terms, it is a workflow that defines how software moves from code to production in an automated and repeatable manner.

- Key Benefits of Jenkins Pipeline:
    - Automates CI/CD processes: Streamlines build, test, and release cycles.
    - Better Visibility: Offers clear insight into the build process steps.
    - Pipeline as Code: Allows version control of the build logic.
    - Supports Complex Workflows: Enables parallel execution, conditional logic, and modular design.

Types of Jenkins Pipelines:

1. Declarative Pipeline:
    - Uses a simpler, structured, and easy-to-read syntax.
    - Defined inside a pipeline {} block organized into stages and steps.
    - Provides better validation and simpler error handling in the Jenkins UI.
2. Scripted Pipeline:
    - Uses Groovy-based syntax for full flexibility and power.
    - Defined inside a node {} block where everything is written as code.
    - Ideal for complex workflows requiring advanced scripting and custom logic.

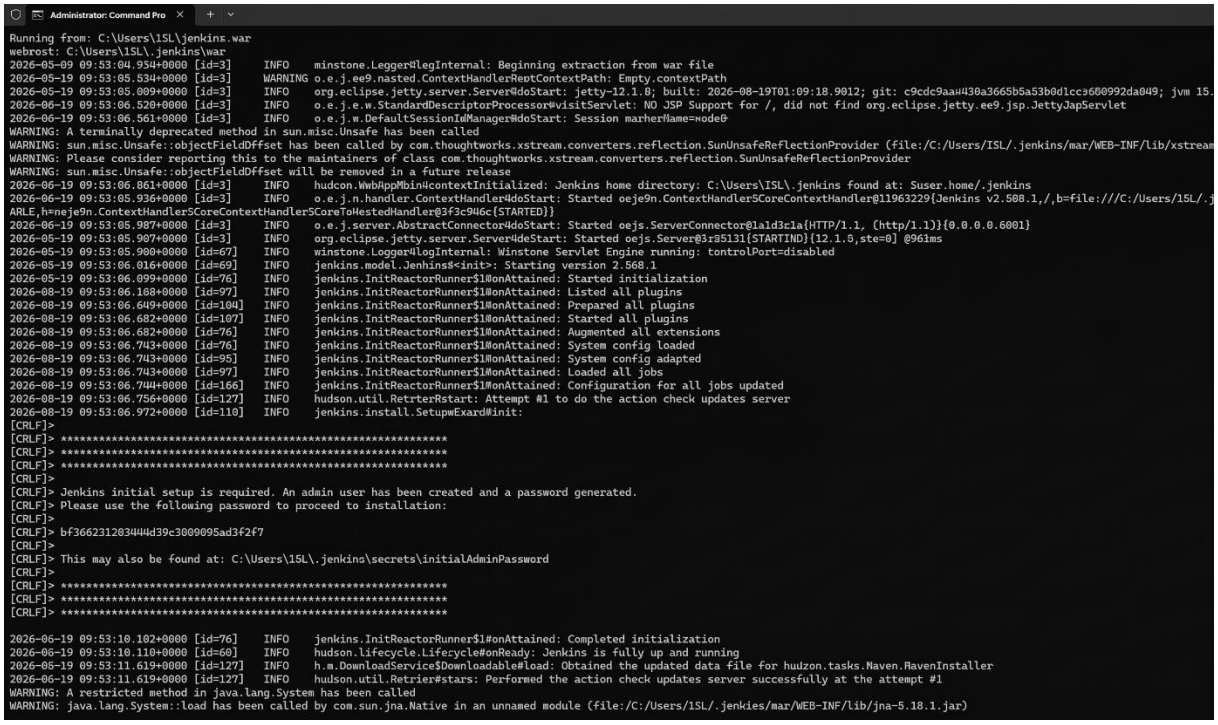
Steps Followed:

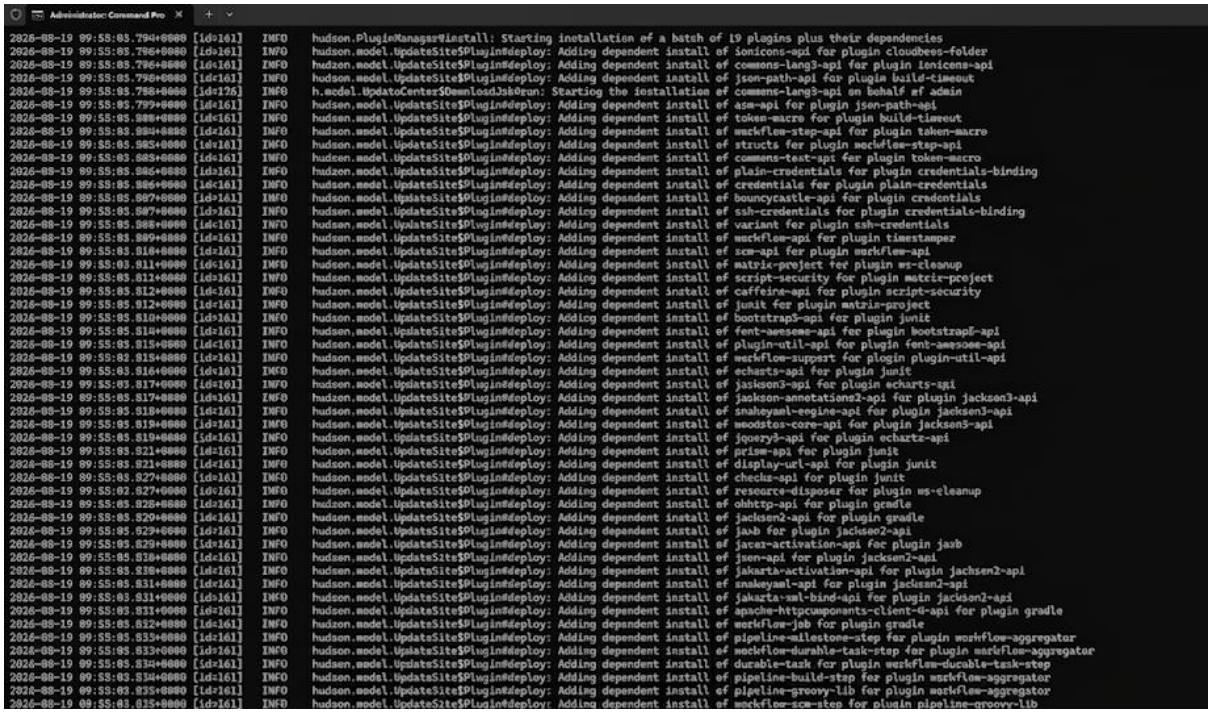
1. Java Verification: Verified that JDK is installed and configured on the system.
2. Download & Launch Jenkins: Downloaded jenkins.war and executed it via the terminal using java -jar jenkins.war.
3. Web Configuration: Accessed http://localhost:8080, entered the generated initialAdminPassword, installed suggested plugins, and configured the Admin account.
4. Pipeline Creation:
    - Created a Declarative Pipeline job executing a basic build step.

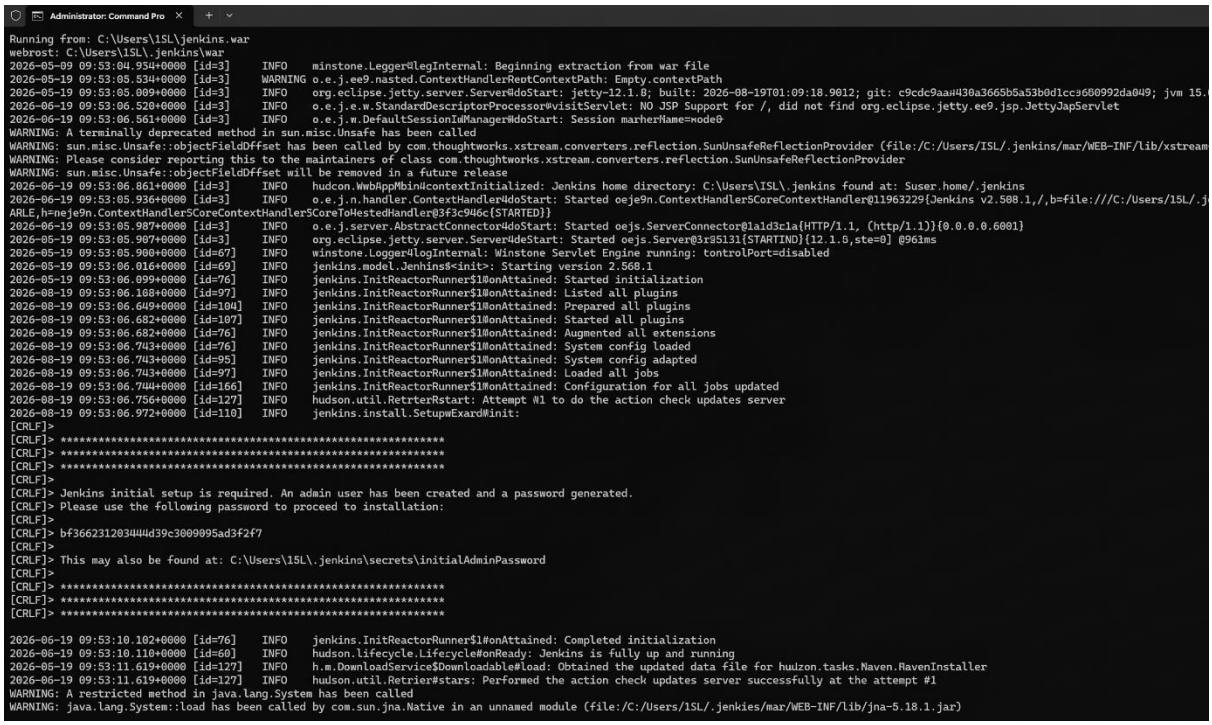
- Created a Scripted Pipeline job using Groovy node syntax.

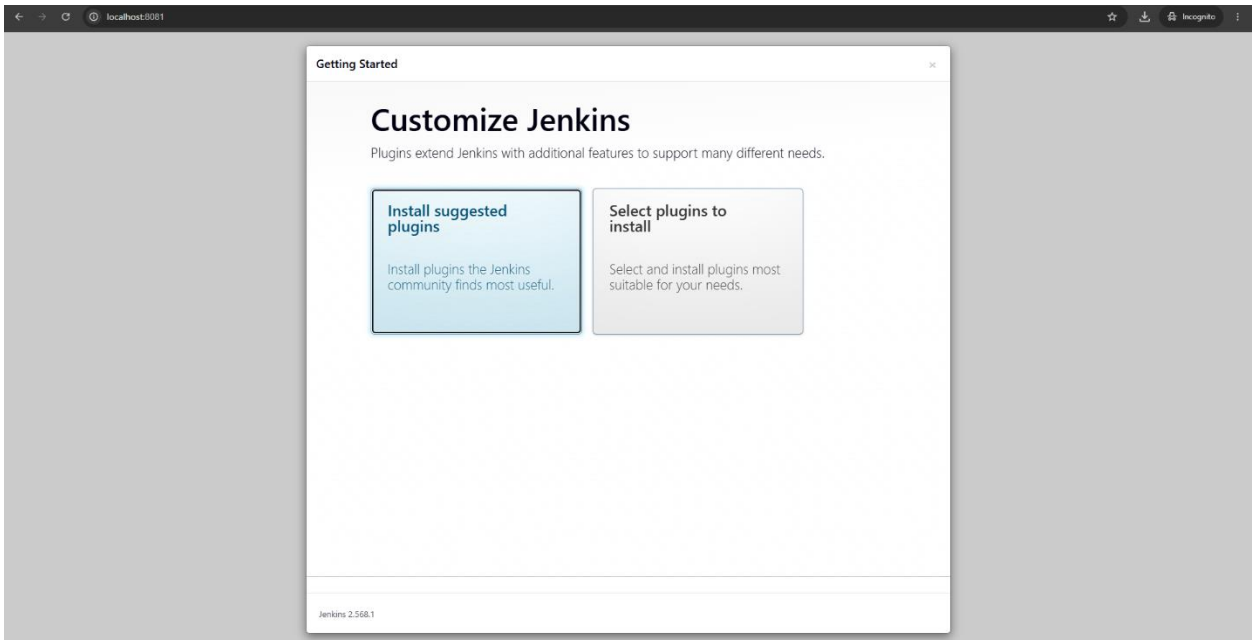
5. Execution: Ran both pipeline jobs and verified successful execution via the Console Output.

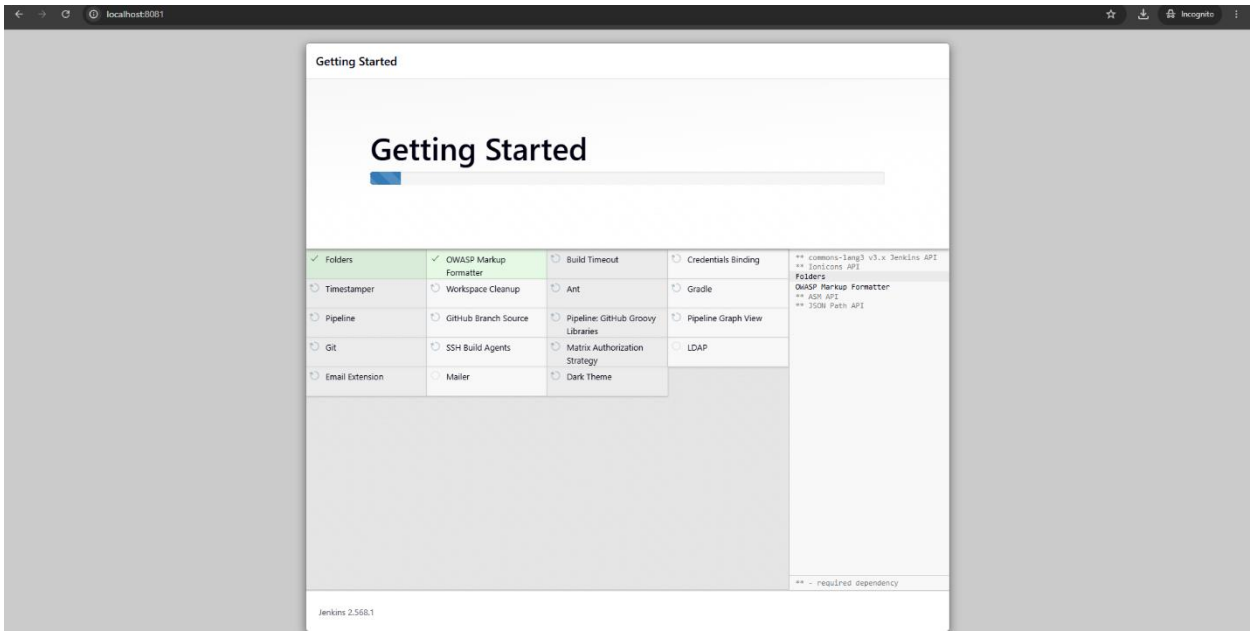
Output:

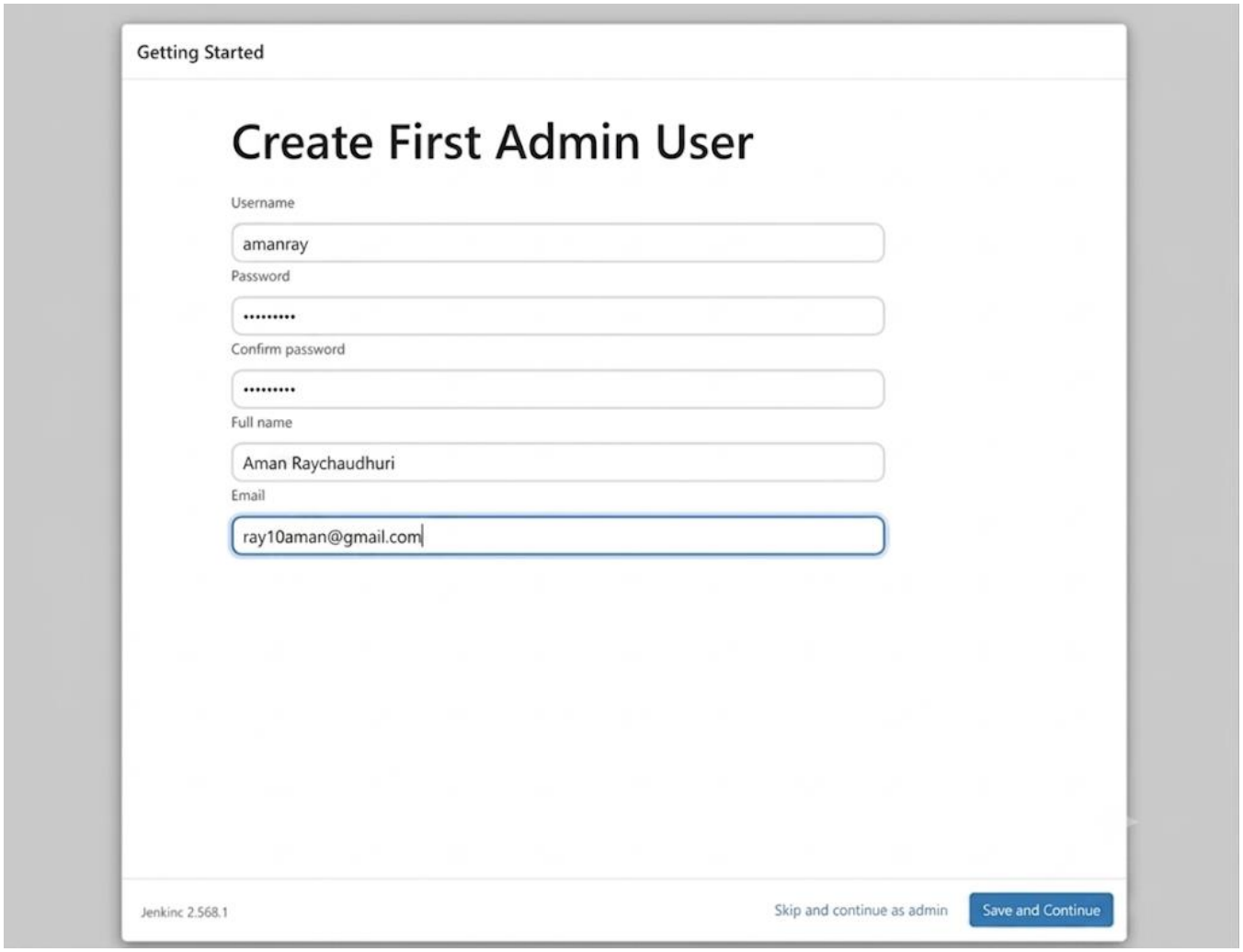


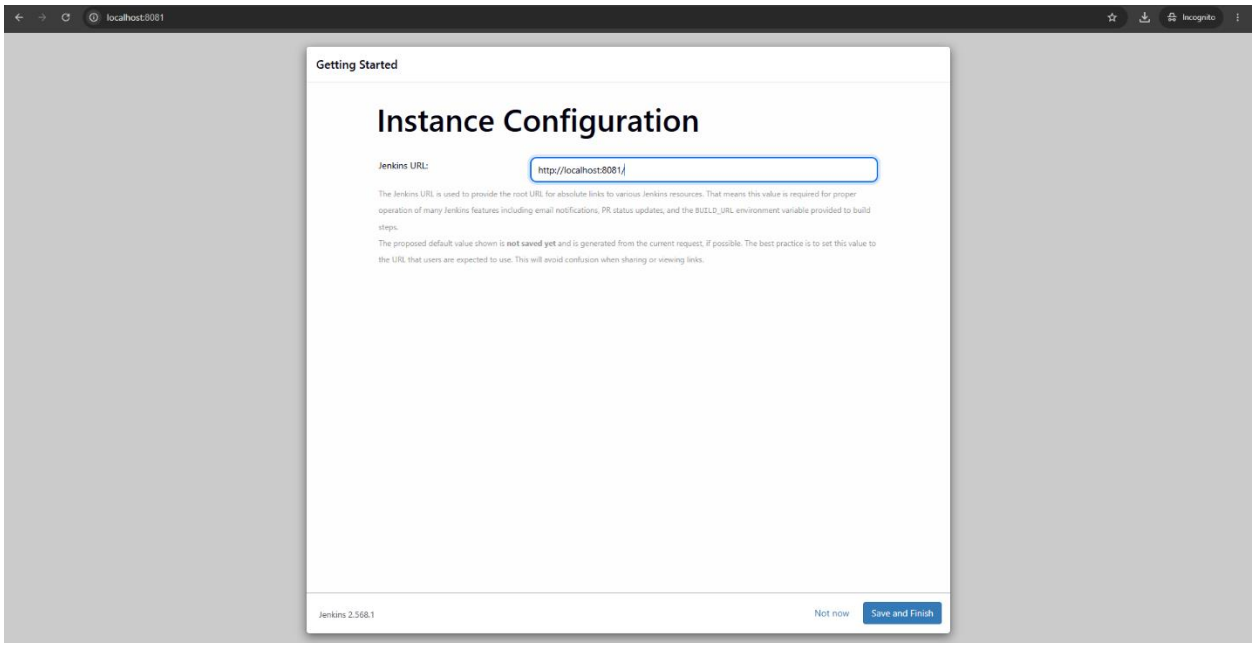


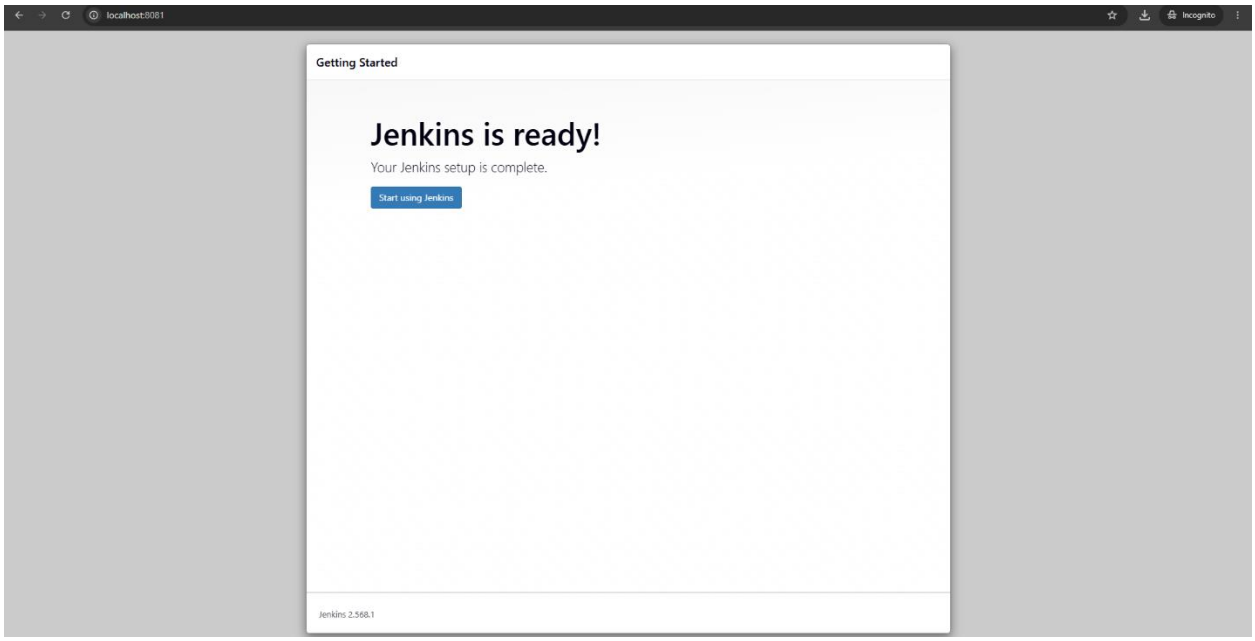

Administrator: Command Pro
Running from: C:\Users\1SL\jenkins.war
webrost: C:\Users\1SL\.jenkins\war
2026-05-09 09:53:04.934+0000 [id=3] INFO minstone.Legger@legInternal: Beginning extraction from war file
2026-05-19 09:53:05.834+0000 [id=3] WARNING o.e.j.ee9.nasted.ContextHandler$eotContextPath: Empty.contextPath
2026-05-19 09:53:05.909+0000 [id=3] INFO org.eclipse.jetty.server.Server@doStart: jetty-12.1.0; built: 2026-08-19T01:09:18.9012; git: c9cdc9aa4430a3665b5a53b0d1cc3600992da049; jvm 15.
2026-06-19 09:53:06.520+0000 [id=3] INFO o.e.j.e.w.StandardDescriptorProcessor@visitServlet: NO JSP Support for /, did not find org.eclipse.jetty.ee9.jsp.JettyJapServlet
2026-06-19 09:53:06.561+0000 [id=3] INFO o.e.j.w.DefaultSessionIdManager@doStart: Session marherName=node0
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldDffset has been called by com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider (file:/C:/Users/ISL/.jenkins/mar/WEB-INF/lib/xstream
WARNING: Please consider reporting this to the maintainers of class com.thoughtworks.xstream.converters.reflection.SunUnsafeReflectionProvider
WARNING: sun.misc.Unsafe::objectFieldDffset will be removed in a future release
2026-06-19 09:53:06.861+0000 [id=3] INFO hudcon.WwbAppMbin#contextInitialized: Jenkins home directory: C:\Users\ISL\.jenkins found at: Suser.home/.jenkins
2026-06-19 09:53:05.936+0000 [id=3] INFO o.e.j.n.handler.ContextHandler4doStart: Started oeje9n.ContextHandler$CoreContextHandler@11963229{Jenkins v2.508.1,/,b=file:///C:/Users/15L/.j
ARLE,h=neje9n.ContextHandler$CoreContextHandler$CoreToHestedHandler@3f3c946c{STARTED}}
2026-06-19 09:53:05.987+0000 [id=3] INFO o.e.j.server.AbstractConnector4doStart: Started oejs.ServerConnector@1a1d3c1a{HTTP/1.1, (http/1.1)}{0.0.0.0.6001}
2026-05-19 09:53:05.907+0000 [id=3] INFO org.eclipse.jetty.server.Server4deStart: Started oejs.Server@3r85131{STARTIND}{12.1.5,ste=0] @961ms
2026-05-19 09:53:05.900+0000 [id=67] INFO winstone.Logger4logInternal: Winstone Servlet Engine running: tontrolPort=disabled
2026-05-19 09:53:06.016+0000 [id=69] INFO jenkins.model.Jenhins$<init>: Starting version 2.568.1
2026-05-19 09:53:06.099+0000 [id=76] INFO jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-19 09:53:06.168+0000 [id=97] INFO jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-19 09:53:06.649+0000 [id=104] INFO jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-19 09:53:06.682+0000 [id=107] INFO jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-19 09:53:06.682+0000 [id=76] INFO jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-19 09:53:06.743+0000 [id=76] INFO jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-19 09:53:06.743+0000 [id=95] INFO jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-19 09:53:06.743+0000 [id=97] INFO jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-19 09:53:06.744+0000 [id=166] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-19 09:53:06.756+0000 [id=127] INFO hudson.util.RetrterRstart: Attempt #1 to do the action check updates server
2026-08-19 09:53:06.972+0000 [id=110] INFO jenkins.install.SetupwExard#init:
[CRLF]>
[CRLF]> **
[CRLF]> **
[CRLF]> **
[CRLF]>
[CRLF]> Jenkins initial setup is required. An admin user has been created and a password generated.
[CRLF]> Please use the following password to proceed to installation:
[CRLF]>
[CRLF]> bf3662312034d39c3009095ad3f2f7
[CRLF]>
[CRLF]> This may also be found at: C:\Users\15L\.jenkins\secrets\initialAdminPassword
[CRLF]>
[CRLF]> **
[CRLF]> **
[CRLF]> **
2026-06-19 09:53:10.102+0000 [id=76] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-06-19 09:53:10.110+0000 [id=60] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
2026-05-19 09:53:11.619+0000 [id=127] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for huuzon.tasks.Naven.RavenInstaller
2026-06-19 09:53:11.619+0000 [id=127] INFO hudson.util.Retrier#stars: Performed the action check updates server successfully at the attempt #1
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by com.sun.jna.Native in an unnamed module (file:/C:/Users/1SL/.jenkies/mar/WEB-INF/lib/jna-5.18.1.jar)

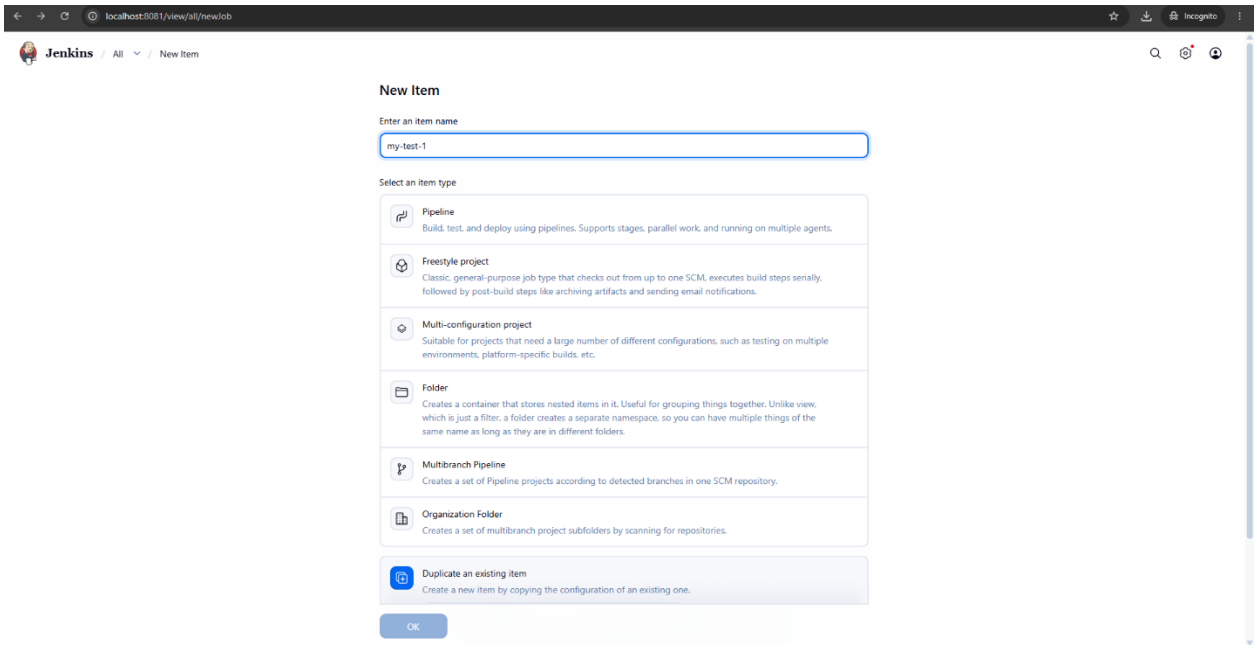

localhost:8081
Incognito
Getting Started
Customize Jenkins
Plugins extend Jenkins with additional features to support many different needs.
Install suggested plugins
Install plugins the Jenkins community finds most useful.
Select plugins to install
Select and install plugins most suitable for your needs.
Jenkins 2.568.1

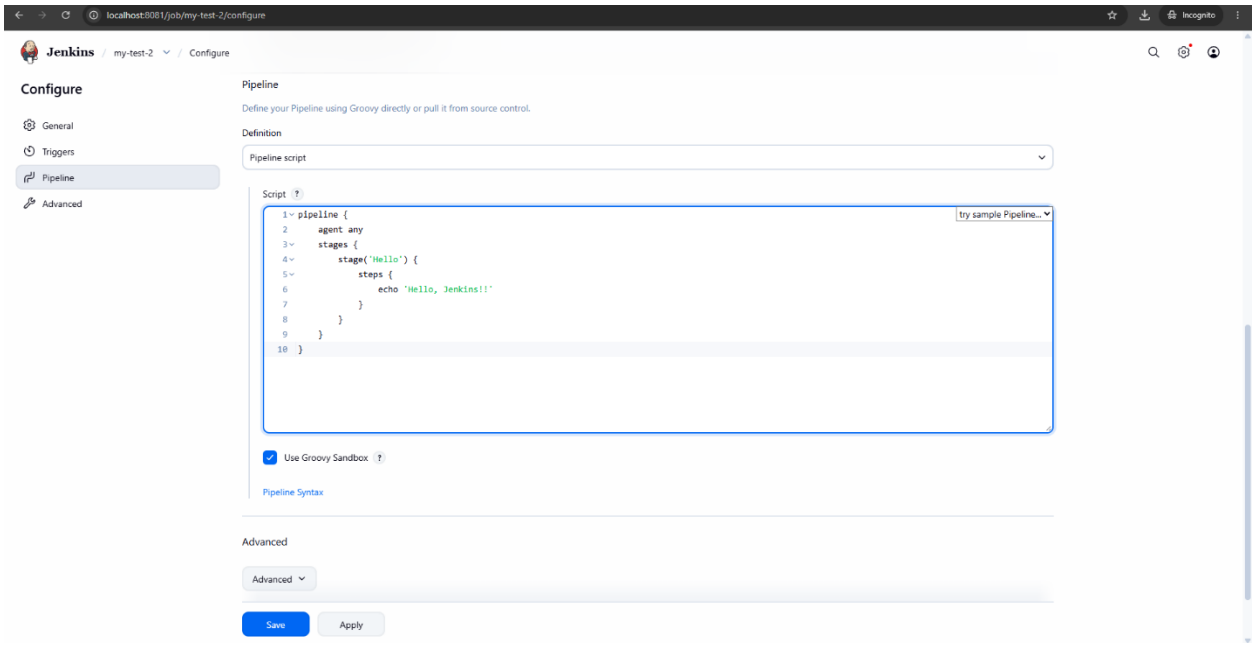

localhost:8081
Incognito
Getting Started
Getting Started
Folders
OWASP Markup Formatter
Build Timeout
Credentials Binding
Timestamper
Workspace Cleanup
Ant
Gradle
Pipeline
GitHub Branch Source
Pipeline: GitHub Groovy Libraries
Pipeline Graph View
Git
SSH Build Agents
Matrix Authorization Strategy
LDAP
Email Extension
Mailer
Dark Theme
** commons-lang3 v3.x Jenkins API
** Ionicons API
Folders
OWASP Markup Formatter
** ASM API
** JSON Path API
** - required dependency
Jenkins 2.568.1

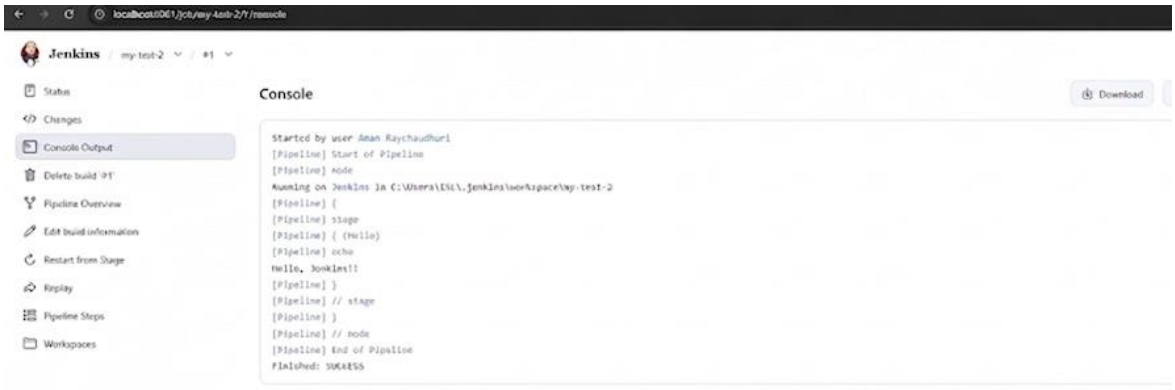

Getting Started
Create First Admin User
Username
amanray
Password
Confirm password
Full name
Aman Raychaudhuri
Email
ray10aman@gmail.com
Jenkinc 2.568.1
Skip and continue as admin
Save and Continue


localhost:8081
Incognito
Getting Started
Instance Configuration
Jenkins URL:
http://localhost:8081/
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.
Jenkins 2.568.1
Not now
Save and Finish


localhost:8081
Incognito
Getting Started
Jenkins is ready!
Your Jenkins setup is complete.
Start using Jenkins
Jenkins 2.568.1

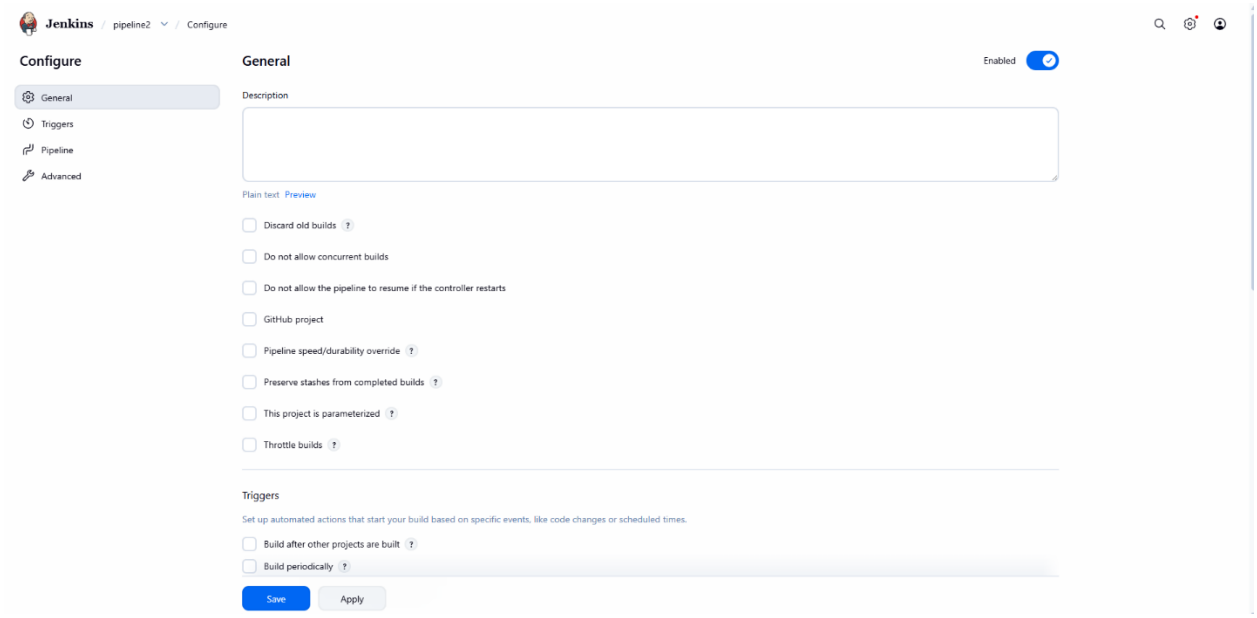

localhost:8081/view/all/newJob
Incognito
Jenkins / All / New Item
New Item
Enter an item name
my-test-1
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
Duplicate an existing item
Create a new item by copying the configuration of an existing one.
OK

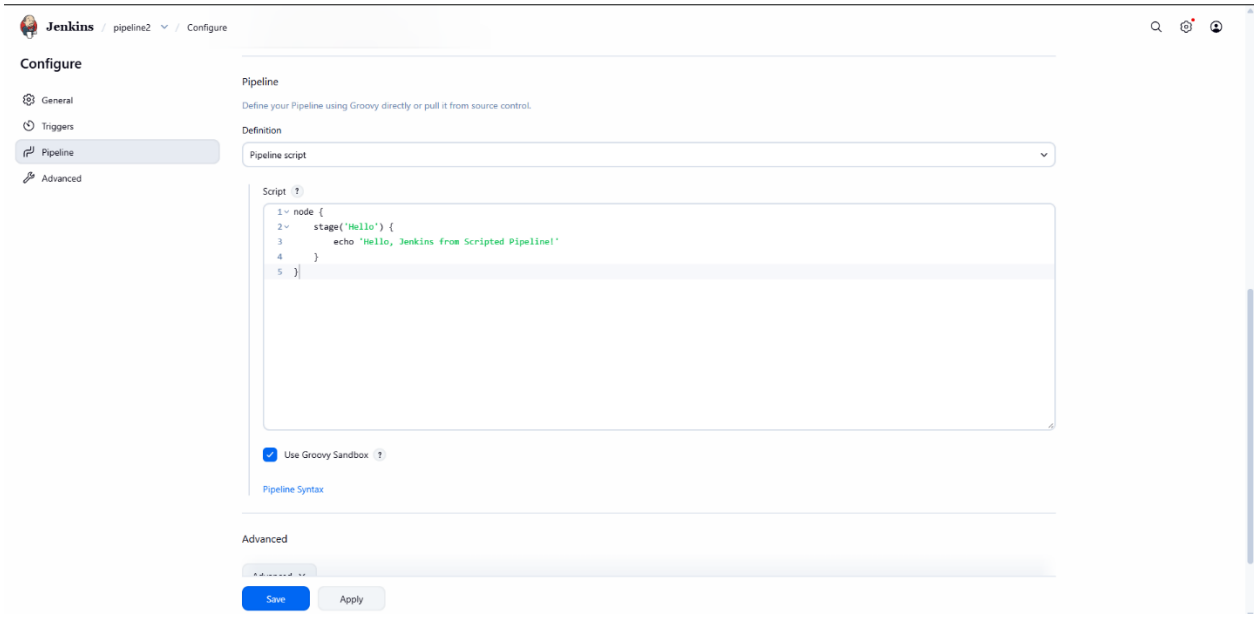

localhost:8081/job/my-test-2/configure
Incognito
Jenkins / my-test-2 / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
try sample Pipeline...
pipeline {
agent any
stages {
stage('Hello') {
steps {
echo 'Hello, Jenkins!!'
}
}
}
}
Use Groovy Sandbox
Pipeline Syntax
Advanced
Advanced
Save
Apply


Jenkins
my-test-2
#1
Status
Changes
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Download
Started by user Aman Raychaudhuri
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\ISI\.jenkins\workspace\my-test-2
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello, Jenkins!!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

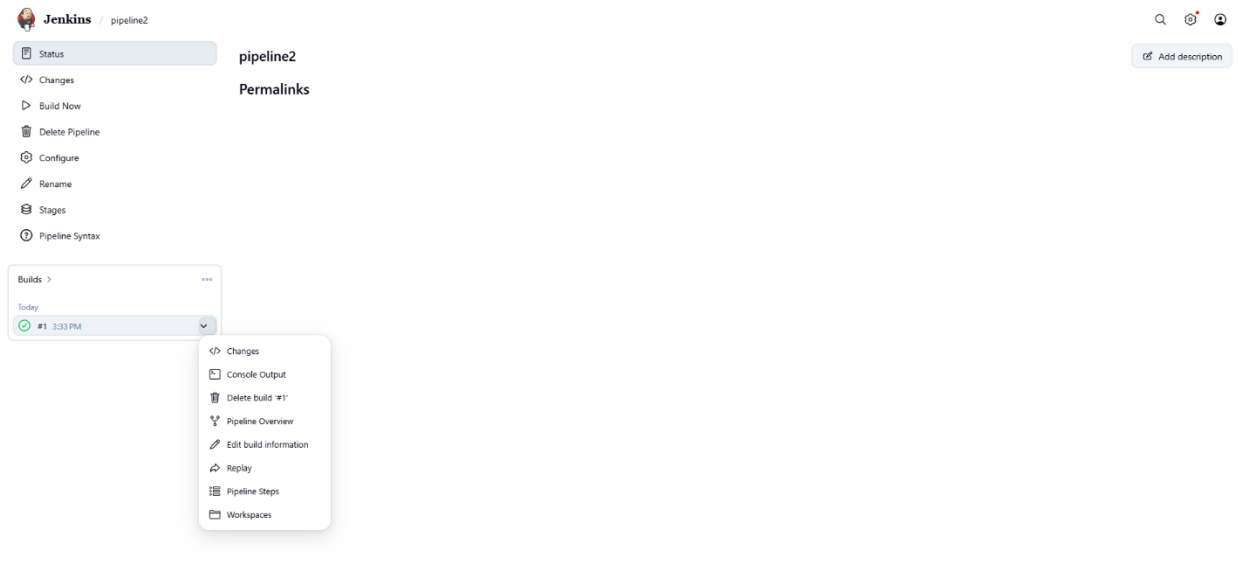

Jenkins
my-test-2
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
Builds
Today
#1 3:30 PM
my-test-2
Add description
Permalinks
REST API
Jenkins 2.568.1

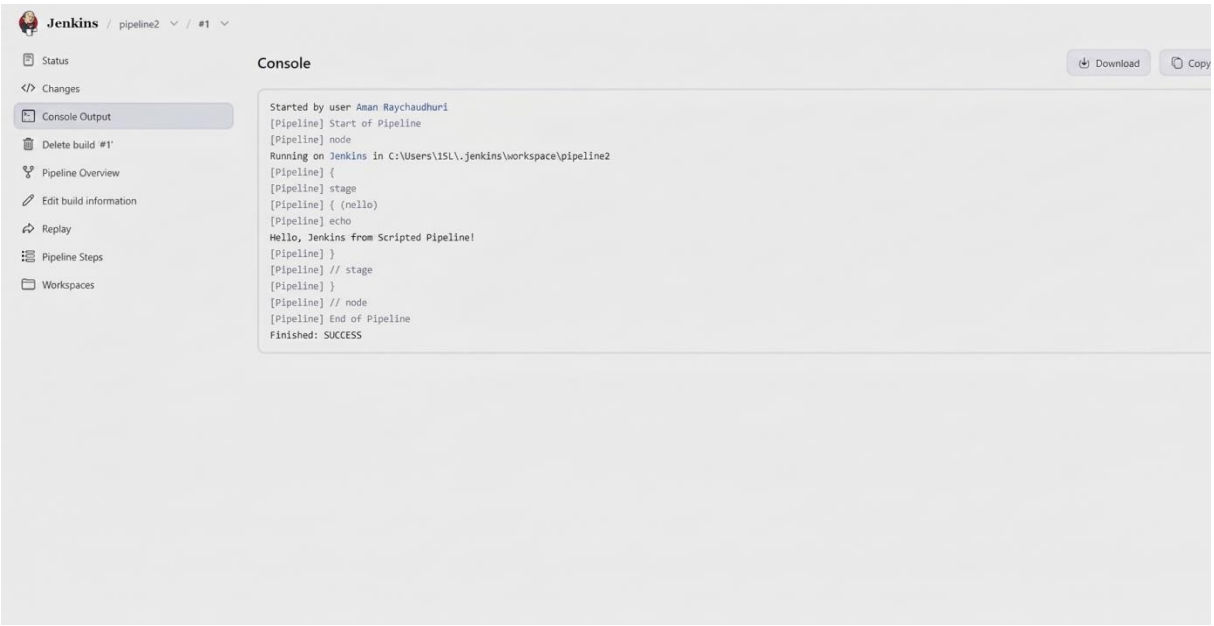

Jenkins / my-test-2
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
my-test-2
Permalinks
Add description
Builds
Today
#1 3:30 PM
Changes
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces


Jenkins / pipeline2 / Configure
Configure
General
Triggers
Pipeline
Advanced
General
Enabled
Description
Plain text Preview
Discard old builds
Do not allow concurrent builds
Do not allow the pipeline to resume if the controller restarts
GitHub project
Pipeline speed/durability override
Preserve stashes from completed builds
This project is parameterized
Throttle builds
Triggers
Set up automated actions that start your build based on specific events, like code changes or scheduled times.
Build after other projects are built
Build periodically
Save
Apply


Jenkins / pipeline2 / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
node {
stage('Hello') {
echo 'Hello, Jenkins from Scripted Pipeline!'
}
}
Use Groovy Sandbox
Pipeline Syntax
Advanced
Save
Apply


Jenkins / pipeline2
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
Builds
Today
#1 3:33 PM
pipeline2
Add description
Permalinks





Conclusion:

Through this experiment, we learned the fundamentals of Continuous Integration and successfully installed and configured Jenkins. Furthermore, we explored both Declarative and Scripted pipelines, demonstrating how Jenkins automates repeatable and efficient CI/CD workflows.

LO mapped successfully.